

A1 – LOGICAL ASSUMPTIONS

Assumptions for use of database

- We added surrogate key for LOAN and RESERVE to allow for future query efficiency.
- Assumed that book_copy_purchase_price, loan_fine_amount will be used in future mathematical operations, so we have set their data types as numeric.

Assumptions for how borrowers can make a loan or reservation

- When a borrower has an overdue loan, the borrower_status will be set to 'banned' and the system will automatically prevent them from making any more loans until the book is returned and their status is changed back to 'permitted'.
- Once the overdue loan is returned and the fine is paid for, the borrower_status is automatically updated, allowing them to borrow again.
- As only 1.8% of loans are returned late, we assume that this can be considered a low incidence so the system will be optimized to handle these without excessive overhead.
- We assume that the system will check the borrower's class to determine how many books they can loan at one time based on borrower_class_loan_limit.
- We use numeric data type for this attribute as we assume that we will be making comparisons, for instance, a borrower can continue to loan if they have borrowed less books than their limit.
- The system will customise the length of loan based on the borrower's class by automatically calculating the loan_due_date based on borrower_class_loan_period.
- To figure out whether a book copy is available for loan directly or if the borrower needs to reserve it, the system will check the book_copy_flag. If it is available for borrowing, a loan will be made. Otherwise, a borrower may make a reservation.
- When the book becomes available, the system will sort the reserve_date_time and find the reserve_phone_no of the borrower with the oldest reservation to call them and arrange a loan.
- Once the LOAN has been updated, we will remove this borrower's entry from the RESERVE relation.

To handle book copies that are stolen, lost or damaged, we suggest the following

- Add a `book_copy_status` attribute to the `BOOK_COPY` relation. This will indicate the condition or availability of each copy. This is a mandatory attribute and the book copies can either be available, stolen, lost or damaged.
- Implement a check within the loan process to ensure only copies that are not stolen, lost or damaged can be borrowed.
- If a damaged book is later repaired or a stolen or lost book is returned, we can update the status of the book copy.